practice test1

This is a preview of the published version of the quiz

Started: Apr 7 at 12:23pm

Quiz Instructions

The grade does not count. you have 4 attempts. You can see answers at the end.

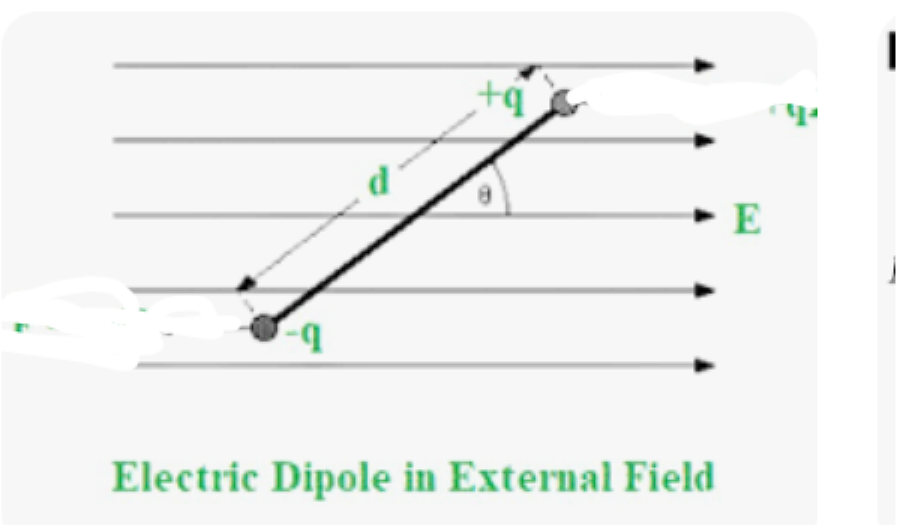
At the end I have added interesting question from a MCAT prep book

Question 1 1 pts

An electric dipole consists of two charges, +2.0 x nC and -2.0 x nC, separated by a distance of 4.0 cm. The dipole is placed in a uniform electric field of 5.0×10^3 N/C, making an angle of 30° with the field direction. Calculate the torque experienced by the dipole. (magnitude)

- $\tau = 2.0 \times 10^{-7}$ Nm
- $\tau = 8.0 \times 10^{-5}$ Nm
- $\tau =$ 3 Nm
- $\tau = 2.0 \times 10^{-6}$ Nm

Question 2 1 pts



This dipole rotates CW or CCW or nothing happens?

IDK

CCW

CW

does not move

Question 3 1 pts

Three point charges are placed along a horizontal line as follows:

- $q_1 = +2.0\,\mu\mathrm{C}$q1=+2.0μC is located at $x = 0.0\,\mathrm{m}$x=0.0m,
- $q_2 = -1.0\,\mu\mathrm{C}$q2=−1.0μC is located at $x = 0.3\,\mathrm{m}$x=0.3m,
- $q_3 = +3.0\,\mu\mathrm{C}$q3=+3.0μC is located at $x = 0.6\,\mathrm{m}$x=0.6m.

Find the net electric field at the point $x = 0.9\,\mathrm{m}$x=0.9m along the line. Assume that positive directions point to the right.

A) $8.1 \times 10^4\ \mathrm{N/C}\ (\mathrm{right})$
B) $5.4 \times 10^4\ \mathrm{N/C}\ (\mathrm{left})$
C) $2.97 \times 10^5\ \mathrm{N/C}\ (\mathrm{right})$
D) $2.7 \times 10^4\ \mathrm{N/C}\ (\mathrm{left})$

D

A

C

B

Question 4 1 pts

Charge $Q_1 = 6.0$ nC is at (0.30 m, 0), charge $Q_2 = -1.0$ nC is at (0, 0.10 m), and charge $Q_3 = 5.0$ nC is at (0, 0). What are the magnitude and direction of the net electrostatic force on the 5.0-nC charge due to the other charges? ($k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9\ \mathrm{N \cdot m^2/C^2}$)

4 × 10-6 N, 26° above -x-axis

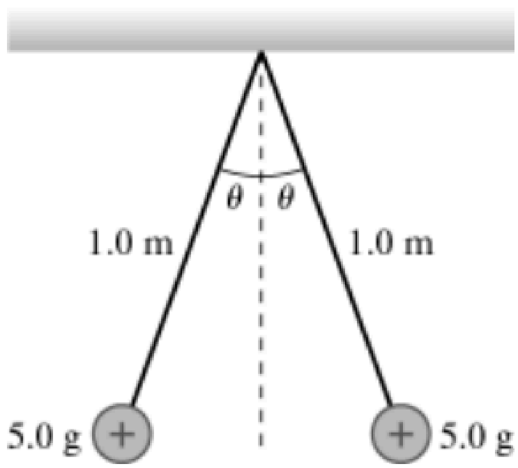
○ HELP

○ 5.4 × 10-6 N, 56° above -x-axis

○ 5.4 × 10-3 N, 34° above -x-axis

Question 5 1 pts

The figure shows two tiny 5.0-g spheres suspended from two very thin 1.0-m-long threads. The spheres repel each other after being charged to +91 nC and hang at rest as shown. What is the angle θ? ($k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9\ \mathrm{N \cdot m^2/C^2}$)



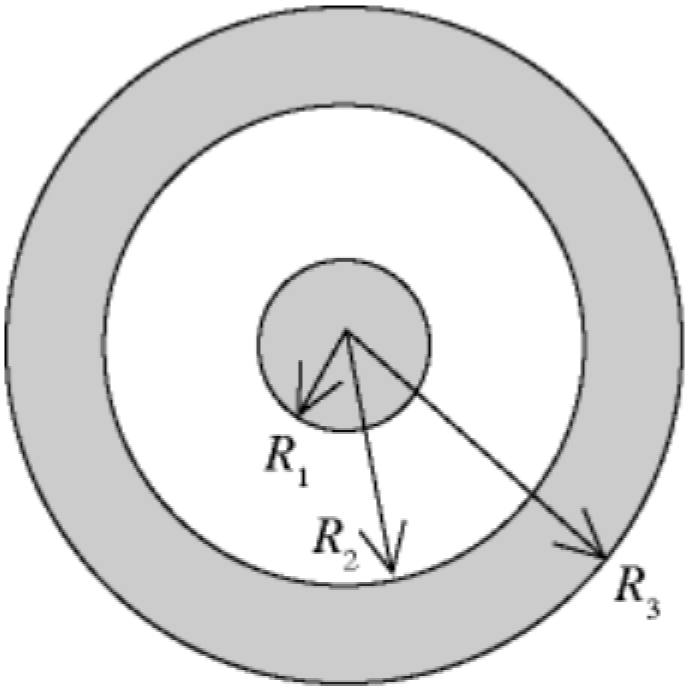
○ 4.1 degrees

○ 12.1 degrees

○ 8.1 degrees

○ 16.1 degrees

Question 6 1 pts



○ -5C, -3C

○ 5C, -5C

○ 3C, 5C

○ -3C, -2C

Question 7 1 pts

A spherical, non-conducting shell of inner radius $r_1 = 10$ cm and outer radius $r_2 = 15$ cm carries a total charge $Q = 15$ μC distributed uniformly throughout the volume of the shell. What is the magnitude of the electric field at a distance $r = 12$ cm from the center of the shell? ($k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9\ \mathrm{N \cdot m^2/C^2}$)

A) 5.75×10^3 N/C
B) zero
C) 2.87×10^6 N/C
D) 5.75×10^6 N/C
E) 2.87×10^3 N/C

○ E

○ C

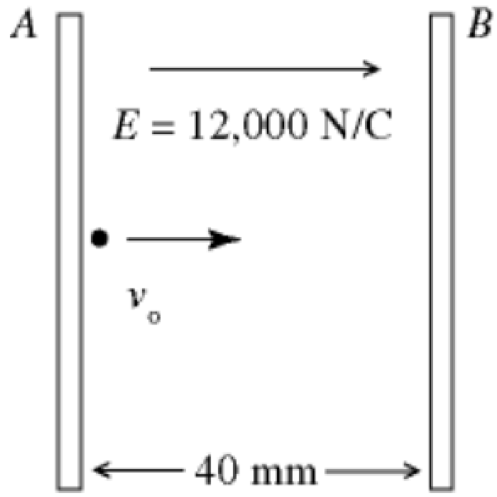
A

D

B

Question 8 1 pts

) A pair of charged conducting plates produces a uniform field of 12,000 N/C, directed to the right, between the plates. The separation of the plates is 40 mm. An electron is projected from plate A, directly toward plate B, with an initial velocity of $v_0 = 1.0 \times 10^7$ m/s, as shown in the figure. ($e = 1.60 \times 10^{-19}$ C, $\varepsilon_0 = 8.85 \times 10^{-12}$ $C^2/N \cdot m^2$, $m_{el} = 9.11 \times 10^{-31}$ kg) The distance of closest approach of the electron to plate B is nearest to



A) 16 mm.
B) 18 mm.
C) 20 mm.
D) 22 mm.
E) 24 mm.

C

D

B

A

E

Question 9 1 pts

A

C

D

B

Question 10 1 pts

A non-conductive cylinder of infinite length has a uniform volume charge density $\rho = 8.0 \times 10^{-6} \mathrm{C/m^3}$ and a radius $R = 3.0\mathrm{cm}$. Using Gauss's Law, find the magnitude of the electric field inside the cylinder at a distance of $r = 2.0\mathrm{cm}$ from the cylinder's axis.

0

about 5 x 105 N/C

about 9 10^3 N/C

about 8 x 10^3N/C

Question 11 1 pts

A conducting sphere has a uniform surface charge density $\sigma = 2.0 \times 10^{-6} \mathrm{C/m}^2$ and a radius $R = 0.10\mathrm{m}$. Using Gauss's Law, calculate the magnitude of the electric field at the surface of the sphere.

Multiple Choices:

A) $1.13 \times 10^5 \mathrm{N/C}$

B) $2.26 \times 10^5 \mathrm{N/C}$

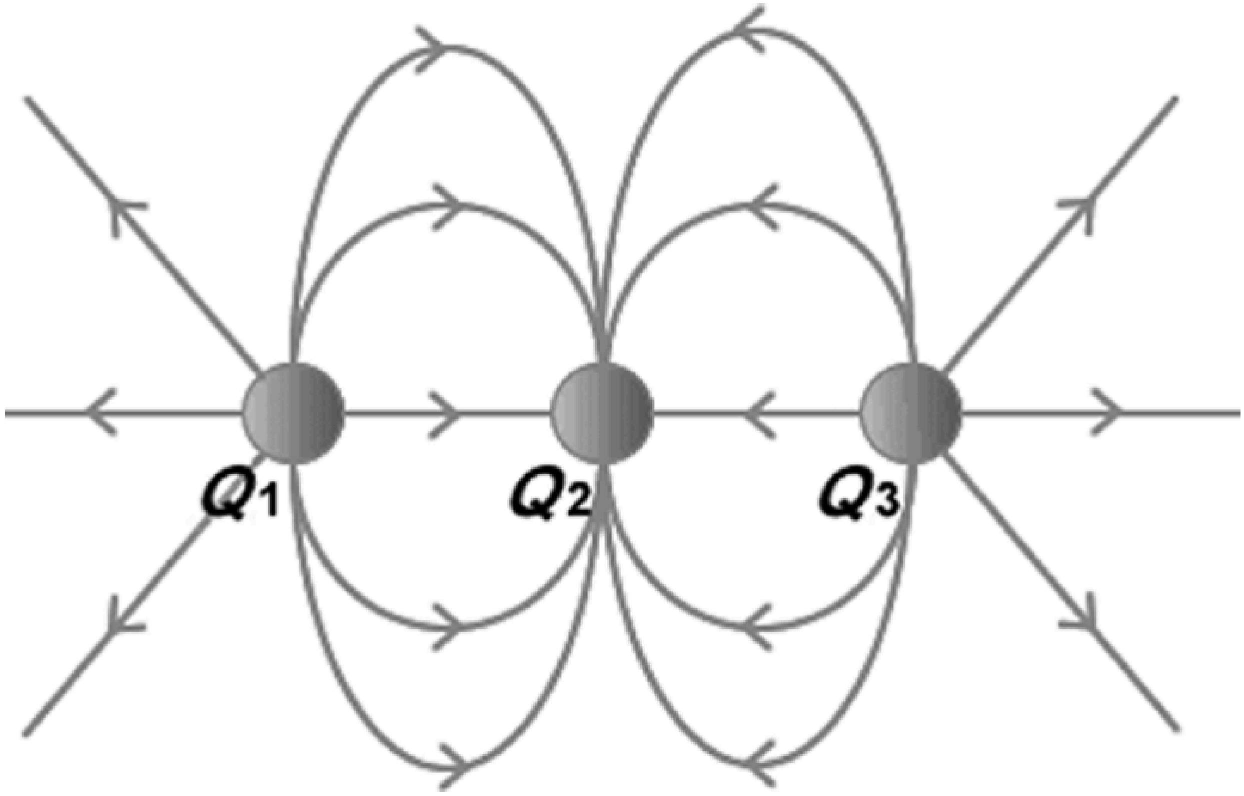
C) $4.52 \times 10^5 \mathrm{N/C}$

D) $9.04 \times 10^5 \mathrm{N/C}$

- B
- D
- A
- C

Question 12 1 pts

The figure shows three electric charges labeled Q_1, Q_2, Q_3, and some electric field lines in the region surrounding the charges. What are the signs of the three charges?



A) Q_1 is positive, Q_2 is negative, Q_3 is positive.
B) Q_1 is negative, Q_2 is positive, Q_3 is negative.
C) Q_1 is positive, Q_2 is positive, Q_3 is negative.
D) All three charges are negative.
E) All three charges are positive.

- D
- C
- E
- B
- A

Question 13 1 pts

A uniform linear charge of 2.0 nC/m is distributed along the x axis from $x = 0$ to $x = 3$ m. Which of the following integrals is correct for the y component of the electric field at $y = 4$ m on the y axis?
hint: So you need to find the expression for dEy = dE cos(angle). Draw the situation. Find the cos(angle) as a function of y (given) and distance between P and dq like I did in class.

a. $\int_0^3 \frac{72dx}{(16+x^2)^{3/2}}$

b. $\int_0^3 \frac{18dx}{(16+x^2)^{3/2}}$

c. $\int_0^3 \frac{72dx}{16+x^2}$

d. $\int_3^0 \frac{18dx}{16+x^2}$

e. none of these

- a
- e
- d
- b
- c

Question 14 1 pts

- C
- B
- A
- D

Question 15 1 pts

Building on the previous question.

F represents the electrostatic force exerted on charged particle S by charged particle R.

If the distance between the centers of the spheres is halved, the magnitude of the force on S due to R will be:

- ○ **A.** $\frac{F}{2}$.
- ○ **B.** $\frac{F}{4}$.
- ○ **C.** $2F$.
- ○ **D.** $4F$.

○ C

○ D

○ A

○ B

Question 16 1 pts

Building on the same questions

R is on left, S is on right with force on S pointing to the left

F represents the electrostatic force exerted on charged particle S by charged particle R.

Assume the direction of **F** is the same direction as the electric field between R and S. If an electron were placed midway between R and S, the resultant electric force on the electron would be:

○ **A.** toward R.

○ **B.** toward S.

○ **C.** upward in the plane of the page.

○ **D.** downward in the plane of the page.

○ B

○ D

○ C

○ A

Question 17 1 pts

If the electric field at a distance *r* away from charge *Q* is $36\ \frac{\mathrm{N}}{\mathrm{C}}$, what is the ratio of the electric fields at *r*, 2*r*, and 3*r*?

- **A.** 9:3:1
- **B.** 36:9:4
- **C.** 36:18:9
- **D.** 36:18:12

C

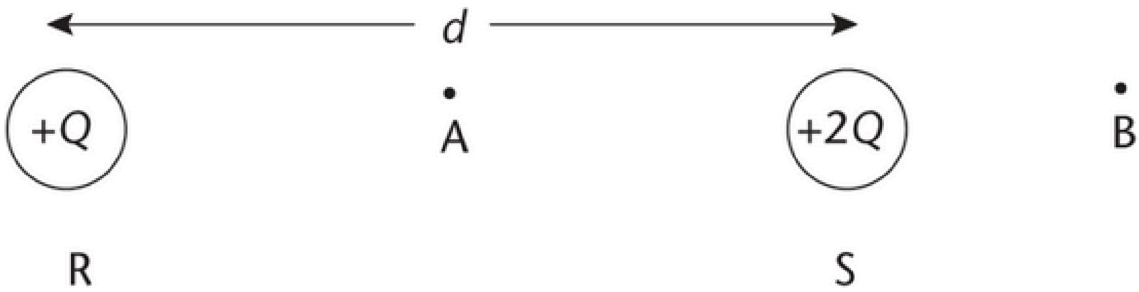
A

D

B

Question 18 1 pts

A positive charge of +Q is fixed at point R a distance d away from another positive charge o +2Q fixed at point S. Point A is located midway between the charges, and point B is a distance $\frac{d}{2}$ from +2Q, as shown below. In which direction will a positive charge move if placed at point A and point B, respectively?



- **A.** Toward the +Q charge for both
- **B.** Toward the +2Q charge for both
- **C.** Toward the +Q charge at point A, and toward the right at point B
- **D.** Toward the +2Q charge at point A, and toward the right at point B

- D
- A
- C
- B

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